

# **$\beta$ -Carboline Alkaloids and Essential Tremor: Exploring the Environmental Determinants of One of the Most Prevalent Neurological Diseases**

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Essential tremor (ET) is among the most prevalent neurological diseases, yet its etiology is not well understood. Susceptibility genotypes undoubtedly underlie many ET cases, although no genes have been identified thus far. Environmental factors are also likely to contribute to the etiology of ET. Harmane (1-methyl-9H-pyrido[3,4- $\beta$ ]indole) is a potent, tremor-producing  $\beta$ -carboline alkaloid, and emerging literature has provided initial links between this neurotoxin and ET. In this report, we review this literature. Two studies, both in New York, have demonstrated higher blood harmane levels in ET cases than controls and, in one study, especially high levels in familial ET cases. Replication studies of populations outside of New York and studies of brain harmane levels in ET have yet to be undertaken. A small number of studies have explored several of the biological correlates of exposure to harmane in ET patients. Studies of the mechanisms of this putative elevation of harmane in ET have explored the role of increased dietary consumption, finding weak evidence of increased exogenous intake in male ET cases, and other studies have found initial evidence that the elevated harmane in ET might be due to a hereditarily reduced capacity to metabolize harmane to harmine (7-methoxy-1-methyl-9H-pyrido[3,4- $\beta$ ]-indole). Studies of harmane and its possible association with ET have been intriguing. Additional studies are needed to establish more definitively whether these toxic exposures are associated with ET and are of etiological importance.